

ZEERBOX · REQUIREMENTS

ZeerBox requirements

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DOCUMENT CONTROL

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Project TRL	TRL 2 Technology concept formulated		
Repository	github.com/BoujeeEnjinia1701/zeerbox		
Source	docs/03-requirements.md @ 5f1047c		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, design point and status against the concept
0.1	2026-09-24	Amish Chadha	Initial scaffold

ZeerBox requirements

These are first-pass requirements for the concept. Targets are proposals for review, not yet validated with users, and will be checked by calculation at TRL 3 and revised after co-design sessions (see ZBX-PRB-001). The status column gives the TRL 2 estimate from ZBX-PRC-001; every value there is an estimate.

The **design point** used throughout is outside air at 35 °C and 30 % RH (wet bulb about 21.5 °C), a full store of 480 kg of mixed vegetables, and 100 kg of fresh produce loaded per day at 32 °C.

ID	Requirement	Target	Verification (TRL 3 or later)	TRL 2 status (estimate)
R1	Storage capacity	20 or more standard 20 kg crates (400 kg or more) on shelves; one adult can walk in and reach every crate without lifting any crate above 1.6 m	Layout model; later loading trial	Met: 24 crates, about 480 kg; top shelf at about 1.45 m
R2	Cooling at the design point	Mean store air temperature 8 K or more below outside air	Psychrometric and heat balance calculation; later logged field data	Met: about 9 K (store about 26 °C)
R3	Humidity for produce	Store RH 85 % or more while cooling, to limit water loss	Calculation; later logged field data	Not met: about 65 to 75 % at the full airflow
R4	Behavior in humid weather	Controller detects outside air too humid to cool (wet-bulb depression under 4 K) and switches to night ventilation or hold, with a visible mode indicator	Controller logic review; later bench test with sensors	Met by design, unverified
R5	Mold and condensation guard	Pad pump stops if store RH stays above 95 % for 60 min; store air changes at least once per hour whenever produce is inside	Controller logic review	Met by design, unverified
R6	Shelf life	Tomato shelf life at least 1.5 times that of ambient storage in the dry season (for example 6 days to 9 days or more)	Literature comparison; later paired storage trial with the partner	Plausible (ZECC data show about 1.8 times), unverified
R7	Solar operation	Fans, pump and controller run from one PV panel with no grid for a full design day: 10 h of cooling plus 2 h of evening ventilation	Energy balance	At risk: about 350 Wh needed, about 350 Wh available from 100 W; no margin
R8	Water use	70 L or less of water per design day, including pad and wall cavity; sump holds at least one day	Water balance	Met: about 55 L; 60 L sump
R9	Low-voltage and safe	All electrical parts at 12 V DC nominal, fused at the battery; fans guarded; door opens from inside without a key	Design review	Met by design
R10	Local build and repair	Walls, roof, door and shelves built by a local mason and carpenter; every fan, pump, sensor and pad replaceable in 30 min or less with generic parts	Design review; parts availability survey with the partner	Met by design, unverified
R11	Durability	Structure 10 years or more; pad 3 years or more with cleaning; fans and pump 3 years or more in dust	Material review; supplier data	Unverified
R12	Affordable	Complete store, including structure, \$300 or less in parts (from <code>project.yaml</code> budget)	Priced BOM (<code>bom/</code> <code>bom.csv</code>)	Not met: about \$575 in total; the equipment kit alone is about \$235

■ Assumptions

- Direct evaporative pad effectiveness of 75 % for a 100 mm cellulose pad at about 0.6 m/s face velocity. Manufacturer data for deeper pads show 90 % or more ([CELdek specification](#)); 70 to 80 % is assumed for 100 mm.
- The produce and the store fabric together gain about 550 W at the design point (heat balance in ZBX-PRC-001).
- Dry-season solar resource of about 5 peak sun hours, with 30 % combined losses for heat, dust, PWM charging and wiring.
- Tomato shelf life gains are taken from zero energy cool chamber trials, which run cooler and more humid than a forced-air store; R6 may be optimistic.
- Recommended produce conditions follow the UC Davis Postharvest Center (tomato 12.5 to 15 °C for mature green, 90 to 95 % RH). ZeerBox cannot reach these temperatures in the design case; R2 is set on temperature drop, not absolute temperature.

■ Requirements not met at TRL 2

- **R3 (humidity):** forced air at the flow needed to carry 550 W warms by about 2.6 K from supply to exhaust, and RH falls as it warms. Options for TRL 3: wet the inner brick leaf as well as the cavity, cut airflow at night, or add a humidifying wet curtain inside the pad.
- **R7 (solar):** the 100 W panel matches the design-day load with no margin. Options: 150 W panel (about \$20 more), cooling only while the sun is up, or fan speed control.
- **R12 (cost):** the walk-in structure costs more than the whole budget. Options are in `docs/REVIEW.md`, proposed, awaiting Amish.